

The above map indicates the filesystem parameters to mount a filesystem at the 'userhome' folder relative to /home (defined in /etc/auto.master), or in other words, /home/userhome.

Let's test our autofs settings now.

- a) First, check whether the filesystem is manually mounted. If it is, unmount it:

```
[root@rdso ~]# df -h
Filesystem                Size Used Avail Use% Mounted on
/dev/mapper/VolGroup00-LogVol00
                        8.6G  5.3G  3.0G   64% /
/dev/hda1                  99M   13M   82M   14% /boot
tmpfs                      250M    0  250M    0% /dev/shm
```

- b) Let's restart the autofs service:

```
[root@rdso ~]# /etc/init.d/autofs restart
Stopping autounmount:      [ OK ]
Starting autounmount:      [ OK ]
```

- c) Log in as vbg1 user (to auto-mount the NFS exported folder from the server):

```
[root@rdso ~]# su - vbg1
```

- d) Now, check the mounted filesystems:

```
[vbg1@rdso ~]$ df -h
Filesystem                Size Used Avail Use% Mounted on
/dev/mapper/VolGroup00-LogVol00
                        8.6G  5.3G  3.0G   64% /
/dev/hda1                  99M   13M   82M   14% /boot
tmpfs                      250M    0  250M    0% /dev/shm
192.168.122.1:/myusersdata
                        19G   17G   1.6G   92% /home/userhome
[vbg1@rdso ~]$
```

Voila! Your folder has been mounted automatically. An important feature of auto-mount is that unused folders get unmounted automatically, which is why it is great for writing backup scripts. We use it often in our day to day jobs while scripting backup jobs. When the backup starts, the backup command (the rsync/cp command – or any other) gets activated; and the remote folders required by it get automatically mounted on the client. After a period of inactivity (that is, when the backup completes) the automounted folder gets unmounted. The set of commands given below demonstrate what I mean. Let's test auto-unmounting.

- a) Log out of the user vbg1, so that /home/userhome access is no longer needed
- b) Note the current time and the mount points in the client
- c) Wait for some time and try again; then note the mount points

```
[root@rdso ~]# date
Thu Oct 21 07:16:58 IST 2010
[root@rdso ~]# df -h
Filesystem                Size Used Avail Use% Mounted on
/dev/mapper/VolGroup00-LogVol00
                        8.6G  5.3G  3.0G   64% /
/dev/hda1                  99M   13M   82M   14% /boot
tmpfs                      250M    0  250M    0% /dev/shm
192.168.122.1:/myusersdata
                        19G   17G   1.6G   92% /home/userhome

WAIT .....
```

After about five minutes of inactivity, check the mount points.

```
[root@rdso ~]# df -h
Filesystem                Size Used Avail Use% Mounted on
/dev/mapper/VolGroup00-LogVol00
                        8.6G  5.3G  3.0G   64% /
/dev/hda1                  99M   13M   82M   14% /boot
tmpfs                      250M    0  250M    0% /dev/shm
[root@rdso ~]# date
Thu Oct 21 07:21:55 IST 2010
[root@rdso ~]#
```

So auto-mount solves the first problem of having to mount all the partitions, manually, and of then keeping them mounted. But the second issue still remains – of all the user partitions being visible:

```
[root@rdso ~]# df -h
Filesystem                Size Used Avail Use% Mounted on
/dev/mapper/VolGroup00-LogVol00
                        8.6G  5.3G  3.0G   64% /
/dev/hda1                  99M   13M   82M   14% /boot
tmpfs                      250M    0  250M    0% /dev/shm
192.168.122.1:/myusersdata
                        19G   17G   1.6G   92% /home/userhome
```

```
[root@rdso ~]# ls /home/userhome/
rajveer testrpm vbg1 websms
```

```
[root@rdso ~]#
```

To solve the second problem of mounting only the user's home folder (similar to the command `=> mount 192.168.122.1:/myusersdata/vbg1 /home/userhome/vbg1`), let's make a few changes in the auto-mount maps /etc/auto.master and /etc/auto.home =>

```
[root@rdso ~]# cat /etc/auto.master | grep home
/home/userhome /etc/auto.home
[root@rdso ~]#
```